

Analysis of Quantum Vulnerable Bitcoin



Anthony Milton

pq-bitcoin.org

MOTIVATION

How much Bitcoin is immediately quantum vulnerable (QV) and why?

“5M-10M BTC”

“6,262,905 BTC”

?

“About 25%”

“now closer to 20%”

OUTLINE

Breakdown of sources of QV, including over time

- 1. Address reuse, revealed keys/scripts**
- 2. Inherently vulnerable script types**
- 3. Fork exposure, with a focus on Bitcoin Cash (BCH)**

Vulnerability assessment of the top 1000 addresses

Any implications for migration efforts, burn vs steal decision?

~6.51M BTC IS IMMEDIATELY QV

32.7% of total supply

Address Reuse: 4.49M BTC (69%)

Inherently Vulnerable: 1.87M BTC (28.8%)

- P2PK: 1.72M
- P2TR: 0.15M
- P2MS: 69 only!

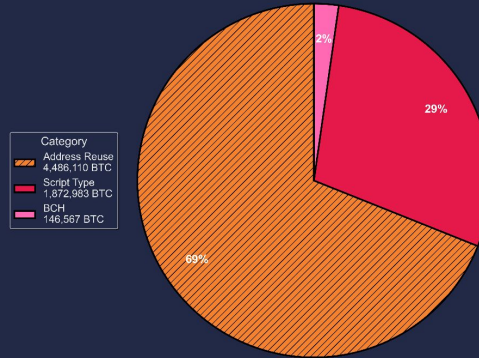
BCH Fork Exposure: 0.15M BTC (2.3%)

**As of block height 900,000 (June 6, 2025)*

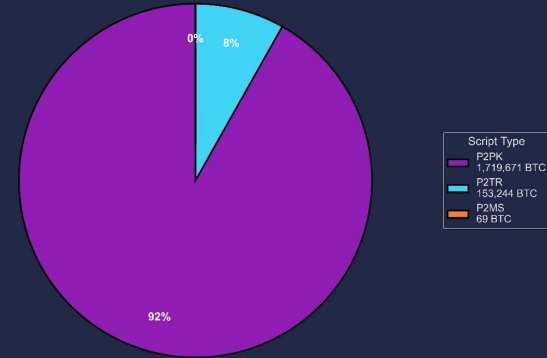
pc-bitcoin.org

Quantum Vulnerability Breakdown at Block 900,000

High-Level Vulnerability Categories



Inherently Vulnerable Script Types



HISTORICAL TRENDS AND BCH FORK EXPOSURE

Reconstruct UTXO set every 50K blocks from block height 400K to 900K

- UTXOs associated with reused addresses
- UTXOs belong to inherently vulnerable scripts
- Generally understand UTXO set composition at snapshots

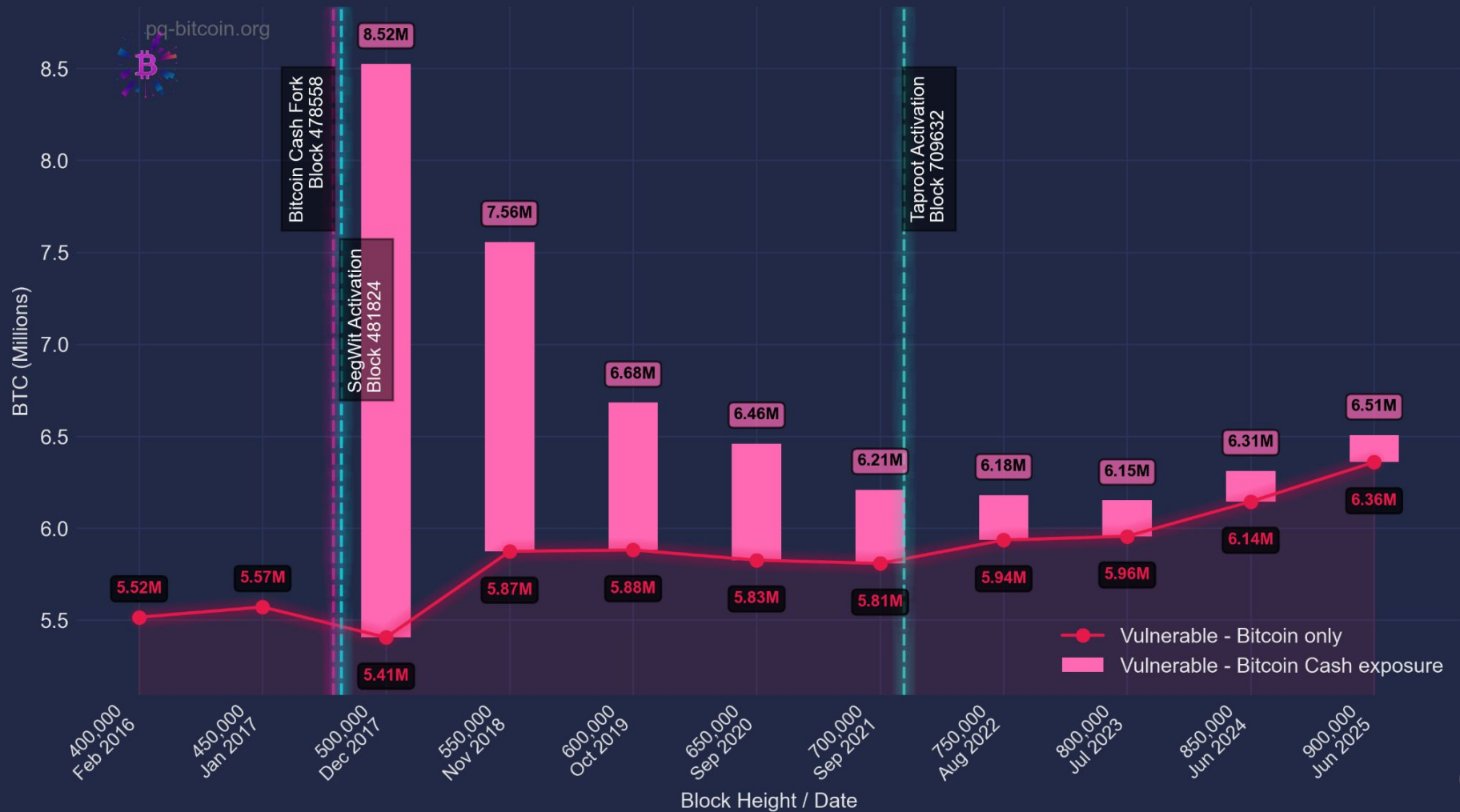
BCH Fork Exposure

- Post fork (478558), determine which TX outputs have been spent on BCH but remain unspent on Bitcoin
- Only consider the marginal contribution of BCH spends

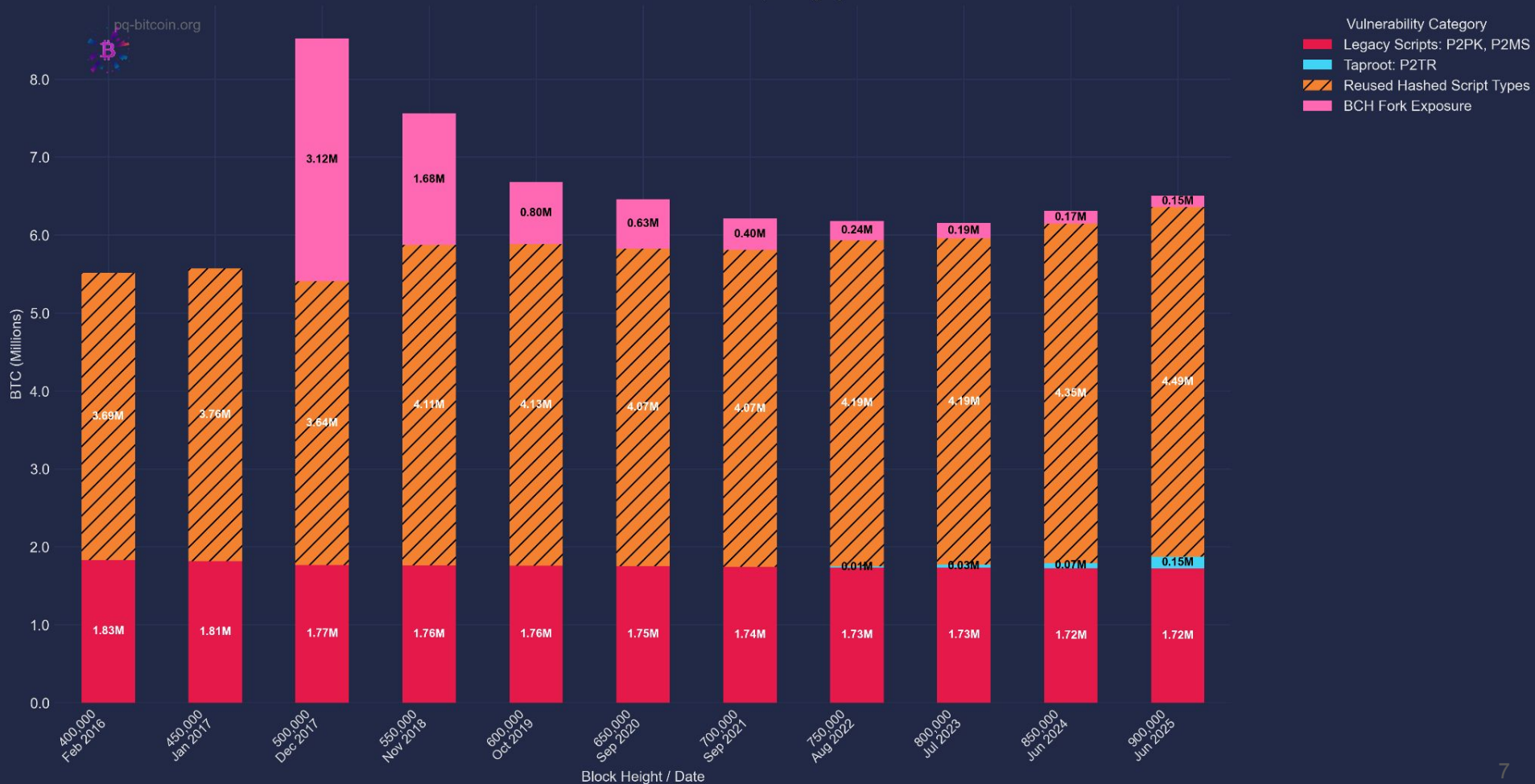


Analysis performed on Google's BigQuery, using [BTC](#) and [BCH](#) datasets

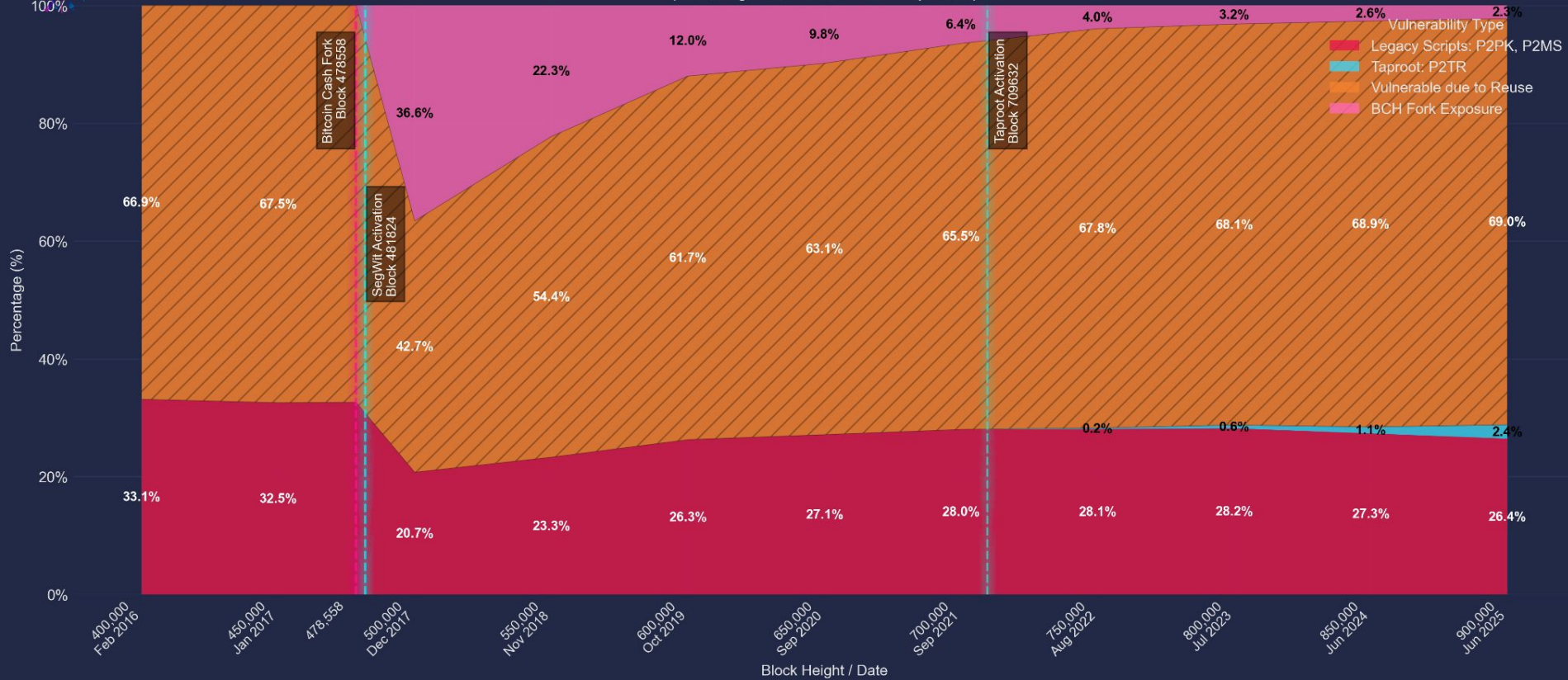
Quantum Vulnerable BTC Over Time



Breakdown of Quantum Vulnerable BTC by Category



Quantum Vulnerability Sources by Percentage (Including Bitcoin Cash Fork Exposure)



ADDRESS REUSE CAVEATS

Not all revealed P2SH/P2WSH scripts are immediately vulnerable, could be protected by non-signature mechanisms:

- Unexpired timelocks - Protection remains until time-lock expires
- Unused hash locks - Only if preimage hasn't been revealed in any prior spend
- Complex conditionals - Where non-hash paths were used

We assume that a majority of P2SH/P2WSH are immediately vulnerable (e.g. they are multisig), and leave analysing the intricacies for future work.

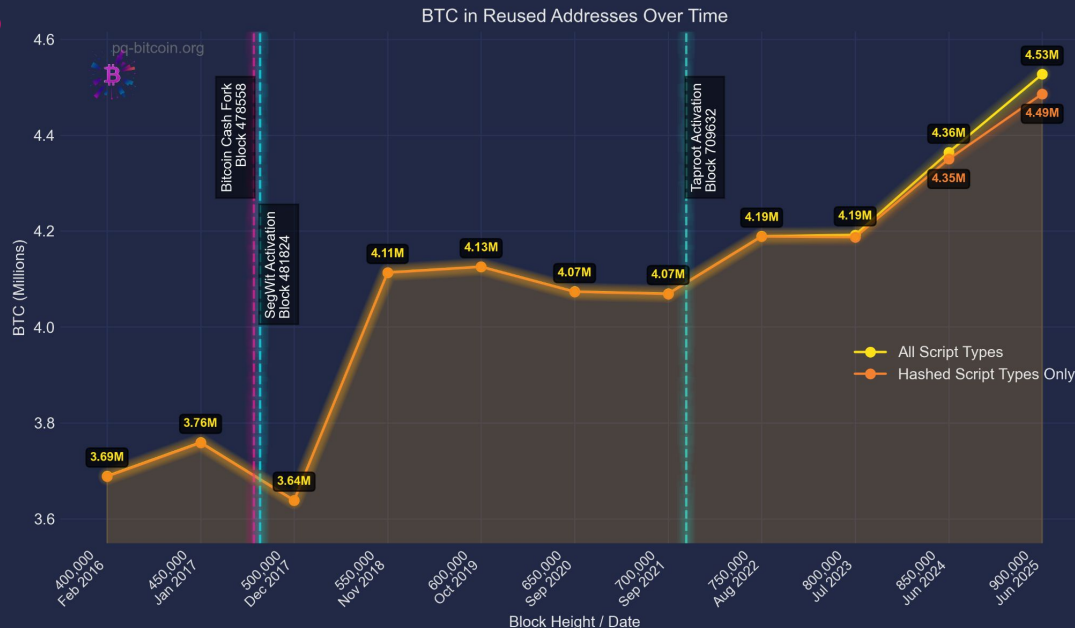
ADDRESS REUSE

500→550K: Dec 2017→Nov 2018

- Bitcoin's first mainstream adoption wave
- Retail and exchanges?

800→900K: Jul 2023→June 2025

- ETFs (Jan 2024) & increased financialization?

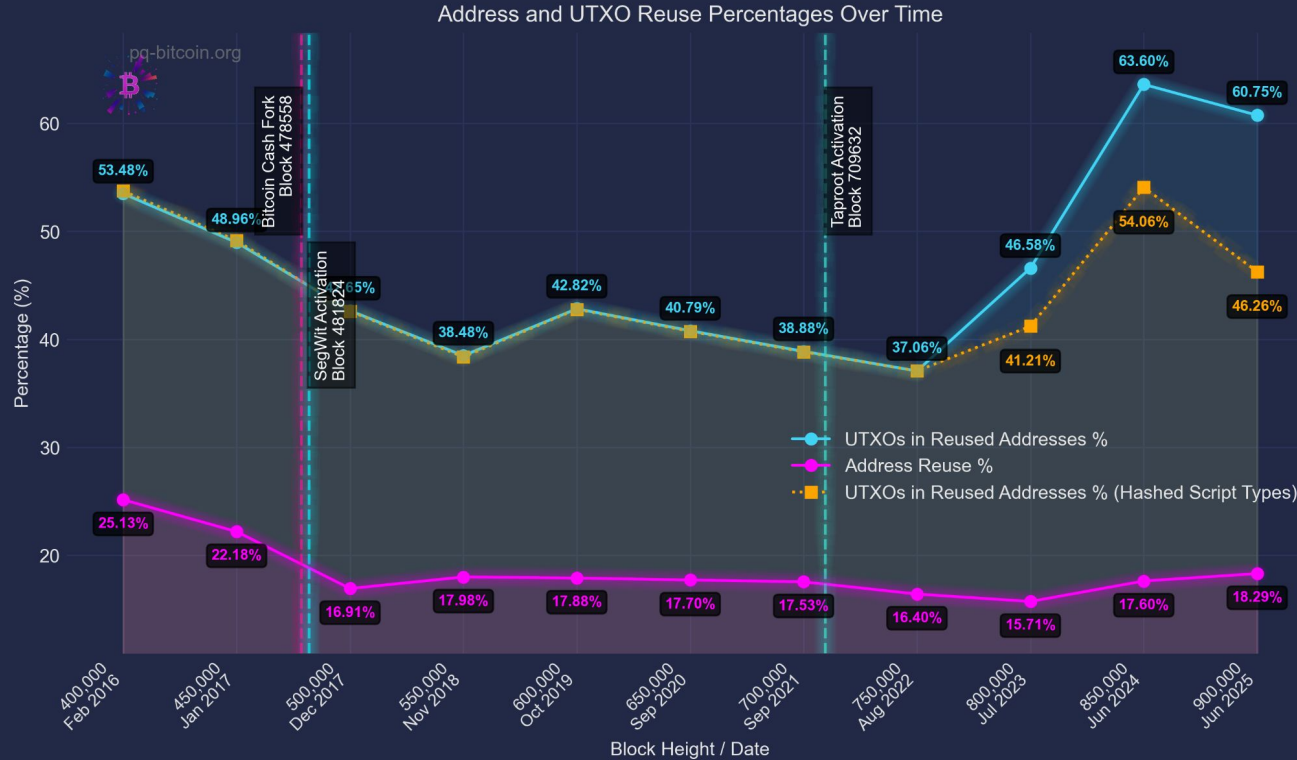


ADDRESS REUSE

Unique address:
addresses that have at
least one UTXO at
each specific block height

Address Reuse % is the
% of unique addresses
that have been reused

**UTXOs in Reused
Addresses %** is the %
of UTXOs that belong
to a reused address



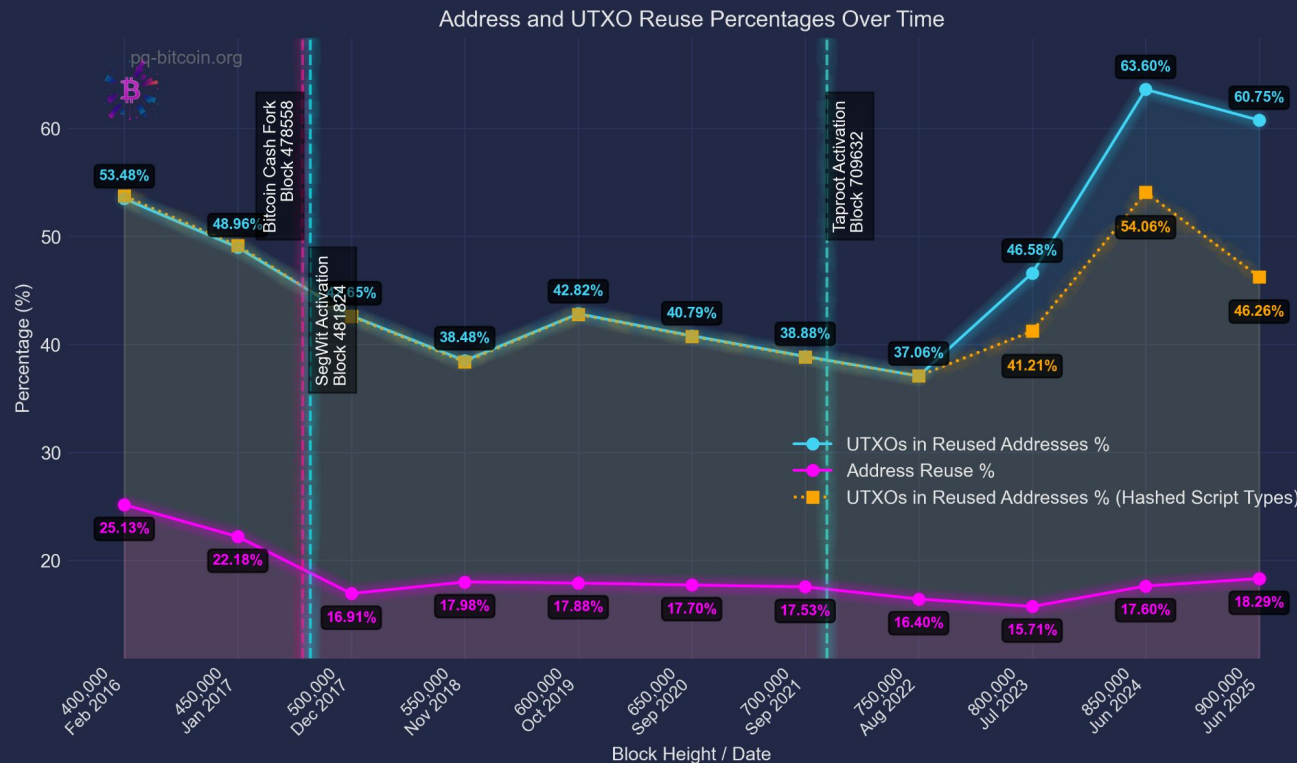
ADDRESS REUSE

Currently:

- < 20% addresses are reused...
- ... but $\frac{2}{3}$ UTXOs belong to reused address

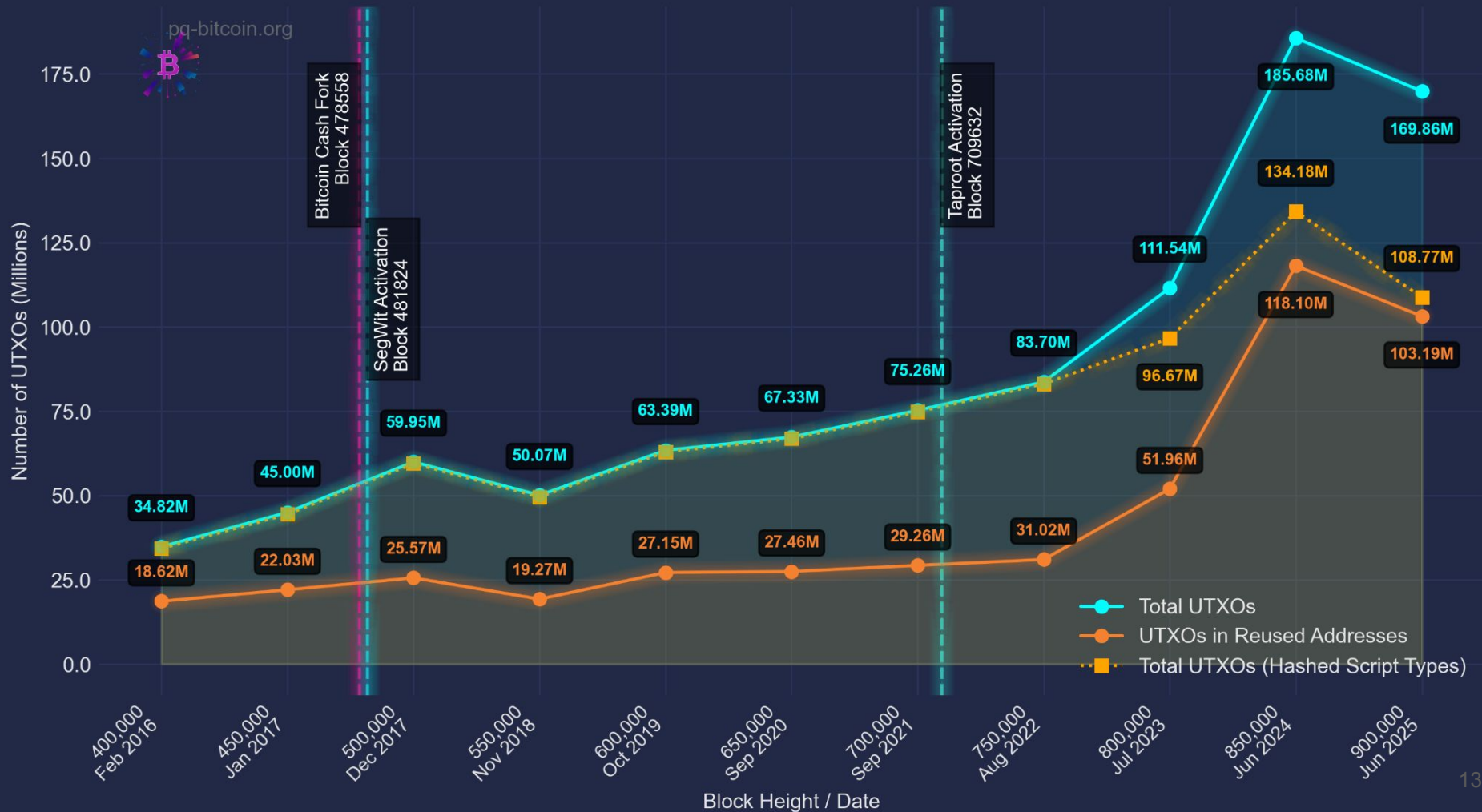
A majority of addresses (80-85%) are single-use

A minority of addresses (15-20%) are reused and account for a majority of UTXOs.



Total UTXOs vs. UTXOs in Reused Addresses Over Time

pq-bitcoin.org



TOP 1000 ADDRESSES ANALYSIS

Top 1000 addresses by balance:

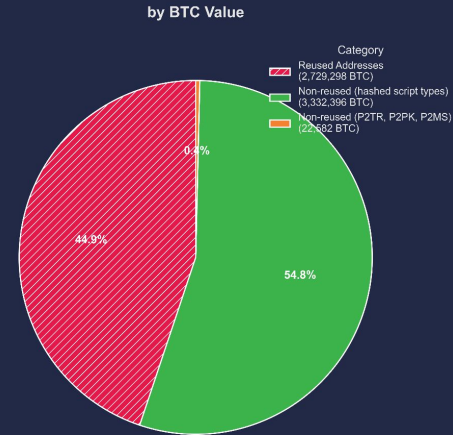
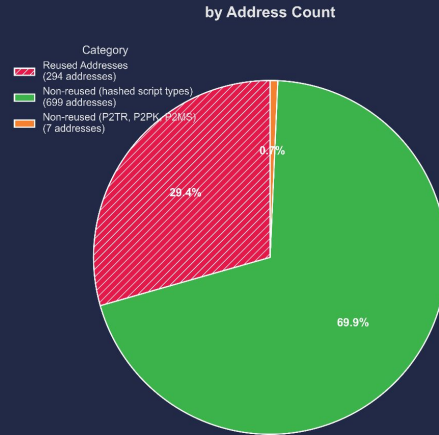
- ~6.08M BTC
- ~31% of total supply
- 1st address: 248,598 BTC (Binance)
- 1000th address: 1,738 BTC (no label)

Reuse:

- 294 addresses
- ~2.73M BTC (~45% of T1000)

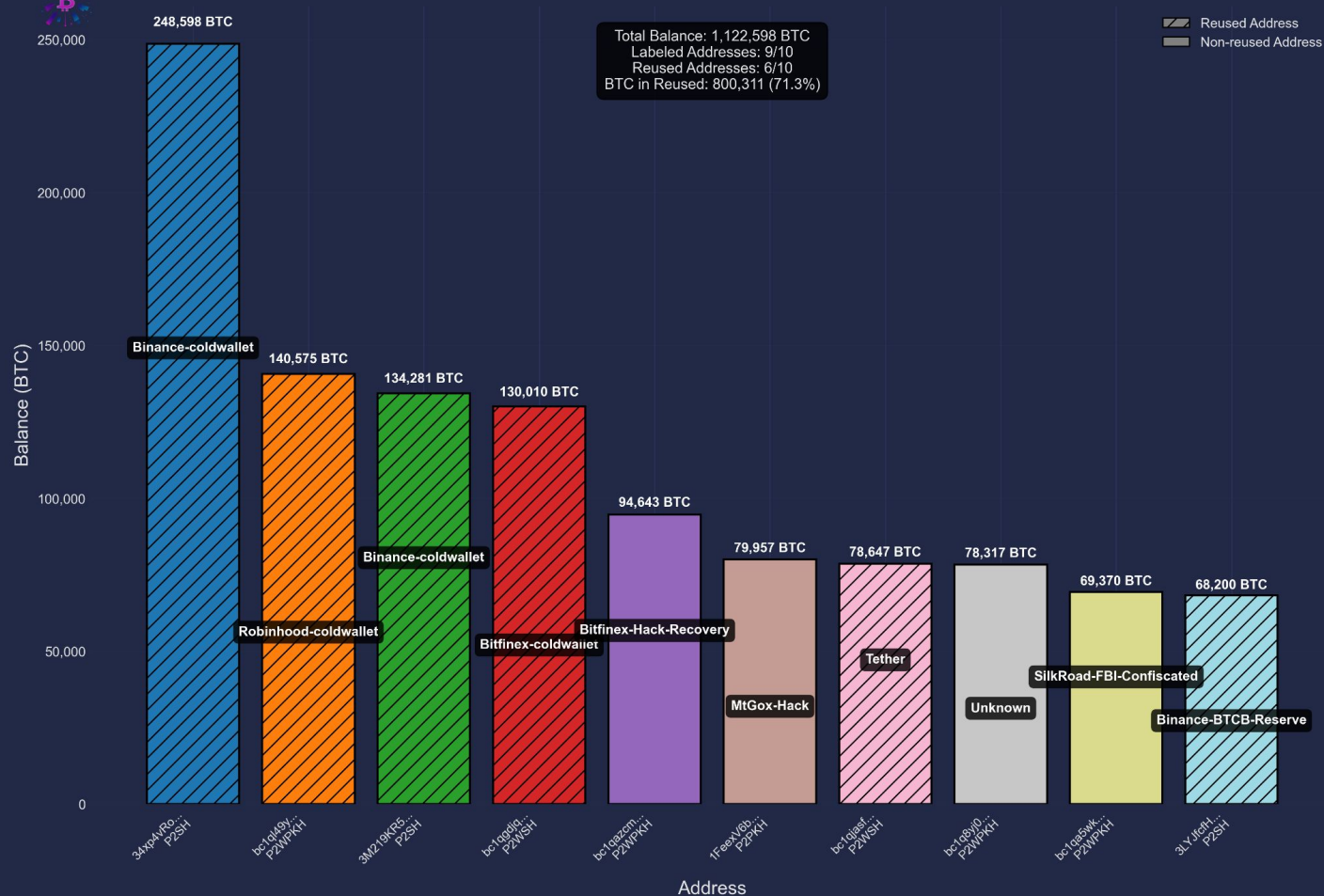
• pq-bitcoin.org

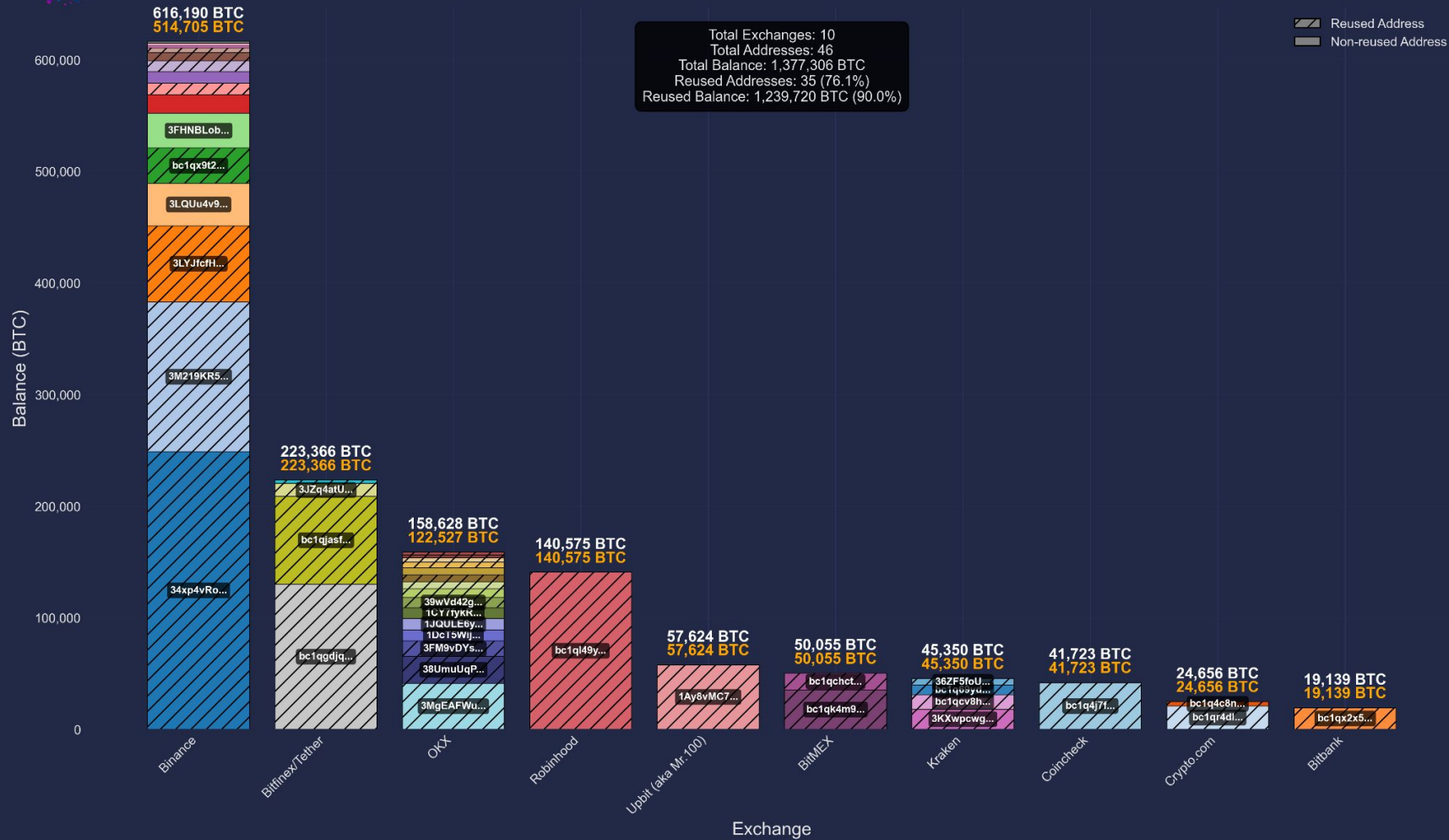
Breakdown of Address Reuse Status in Top 1000 Addresses



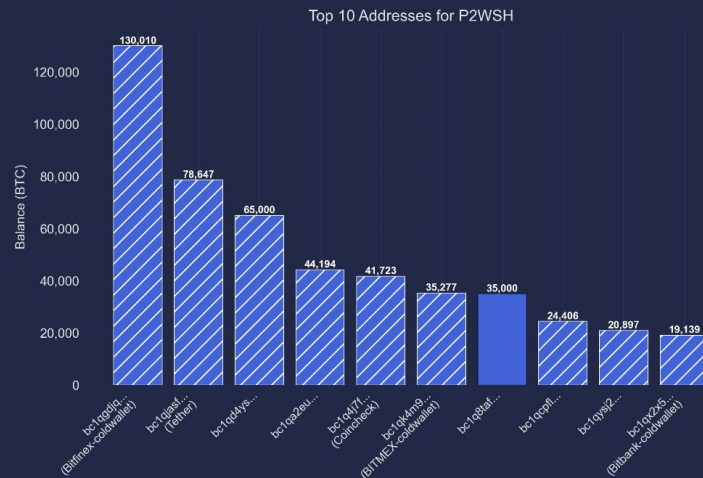
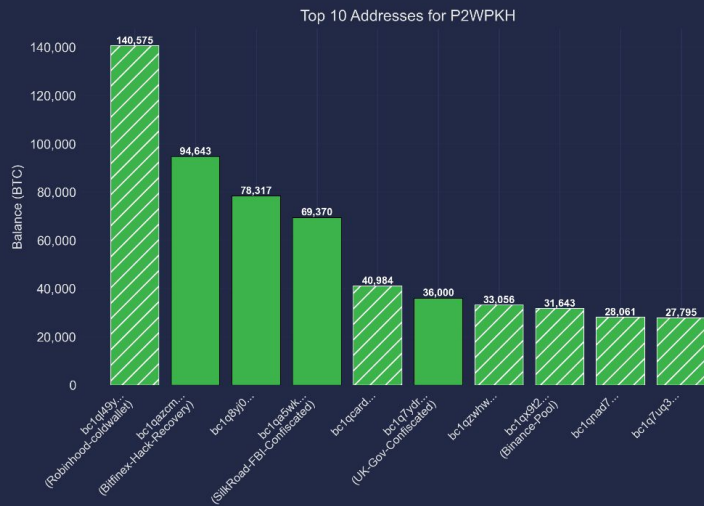
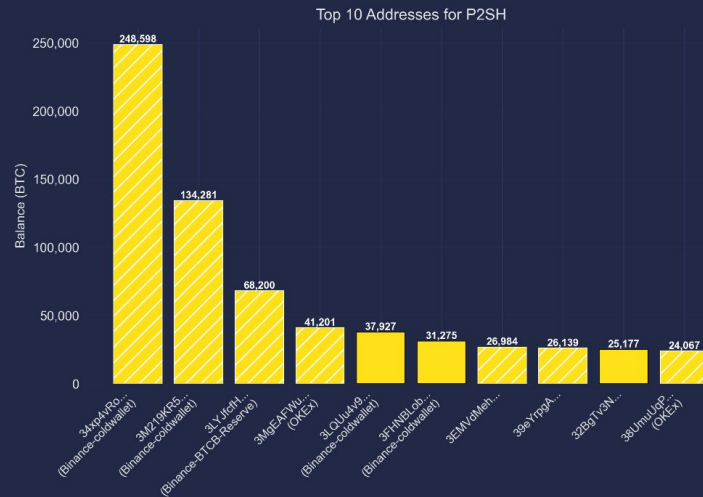
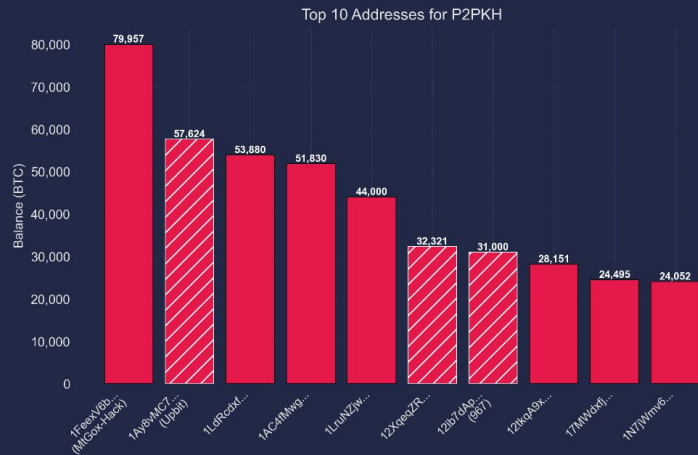
Data sourced from [BitInfoCharts' Rich List](#) (including address labels)

* As of block height ~900,500 (June 10, 2025)



Top 10 Exchange Entities by Balance
(Stacked by Address)

Top 10 Addresses by Hashed Script Type



IMPLICATIONS FOR MIGRATION, BURN/STEAL

We have 3 main sources of immediately QV Bitcoin,
what do they mean for mean for migration, burn/steal

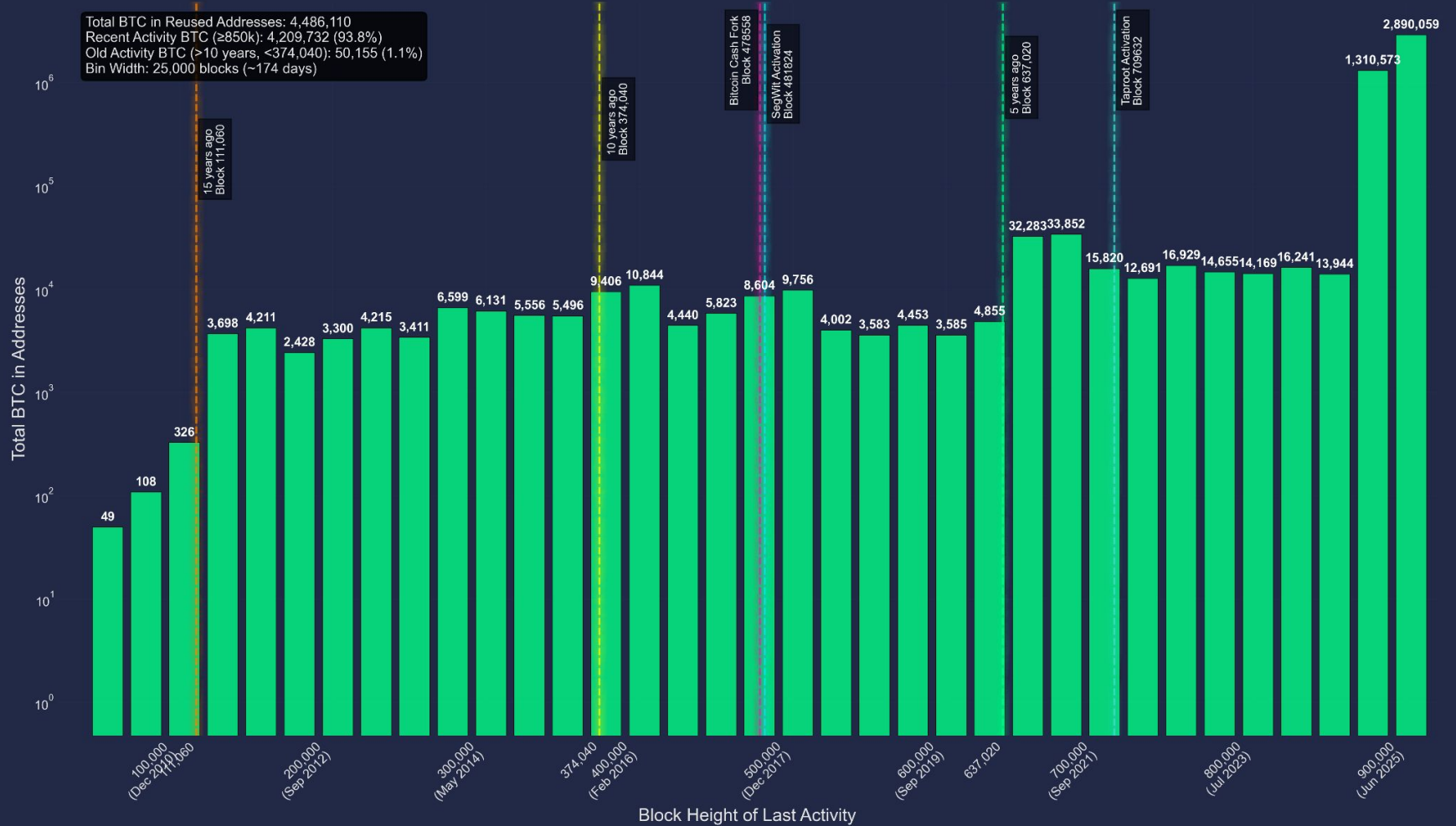
BCH fork exposure:

- Spent on BCH within last 8 years → can still be moved today?
- ∴ Most BCH fork exposure UTXOs can be moved? ~150K BTC

Reused addresses:

- How many reused address UTXOs can be moved?
- Consider last activity of associated addresses

Last Activity Distribution of Reused Bitcoin Addresses (BTC Value)



REUSED ADDRESSES

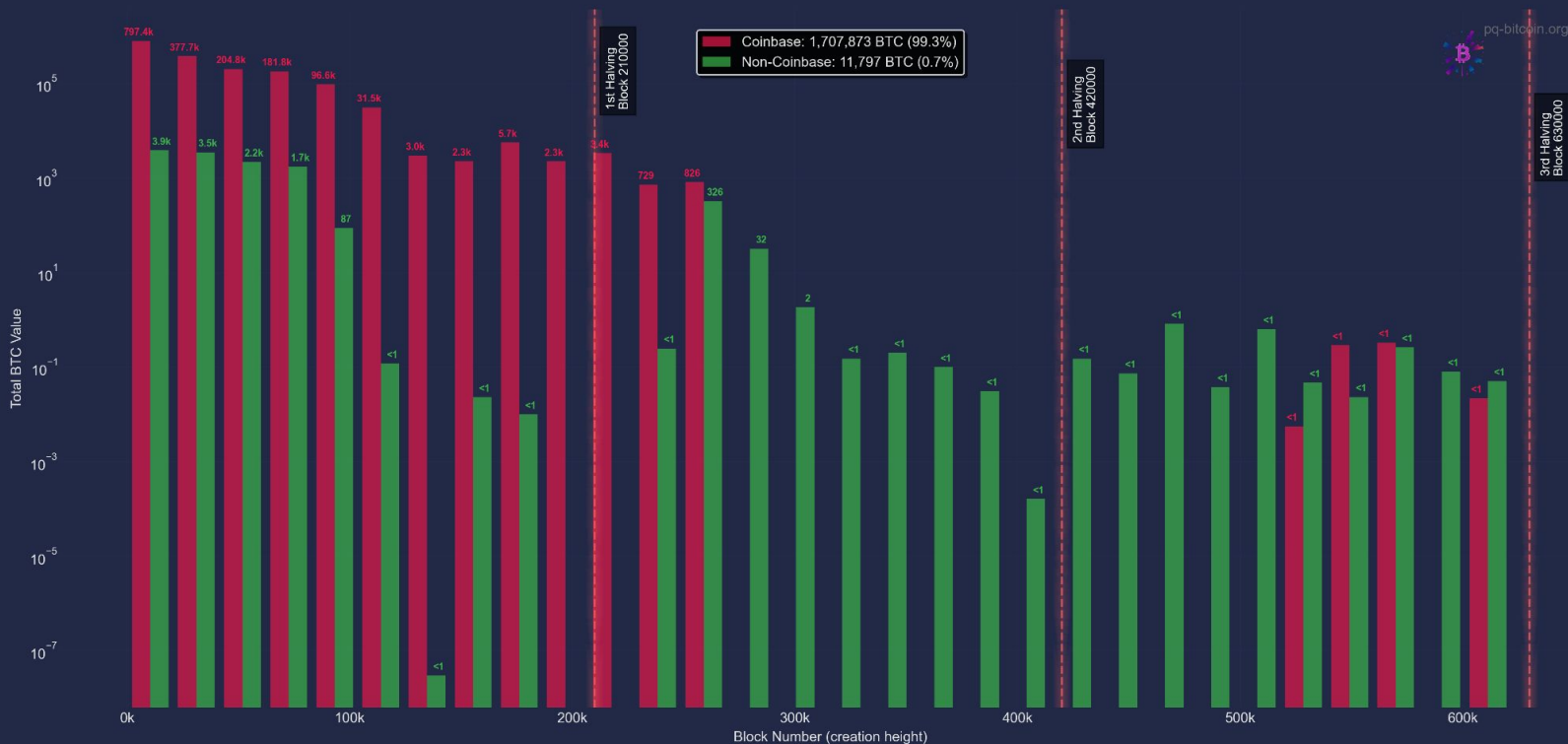
Activity Period	Addresses	Address %	Total BTC	BTC %
Within last 5 years	5354856	63.73	4371782.98	97.45
Between 5 and 10 years	2224323	26.47	64172.36	1.43
Between 10 and 15 years	823216	9.80	49784.88	1.11
More than 15 years	221	0.00	370.20	0.01

Funds associated with reused addresses that we might consider lost (from a last activity perspective) are negligible → can still be moved today?

INHERENTLY VULNERABLE SCRIPTS - P2PK

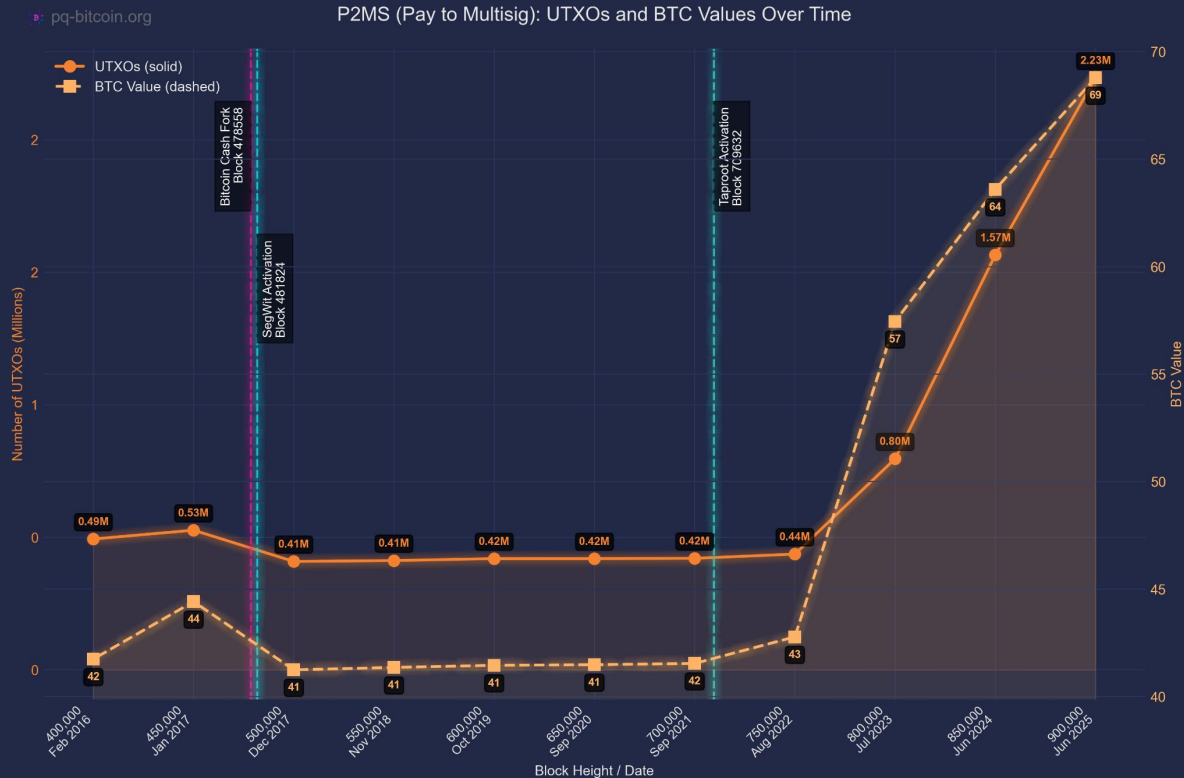
New UTXOs not being created, but significant historical BTC value

BTC Value Distribution by Age: Coinbase vs Non-Coinbase



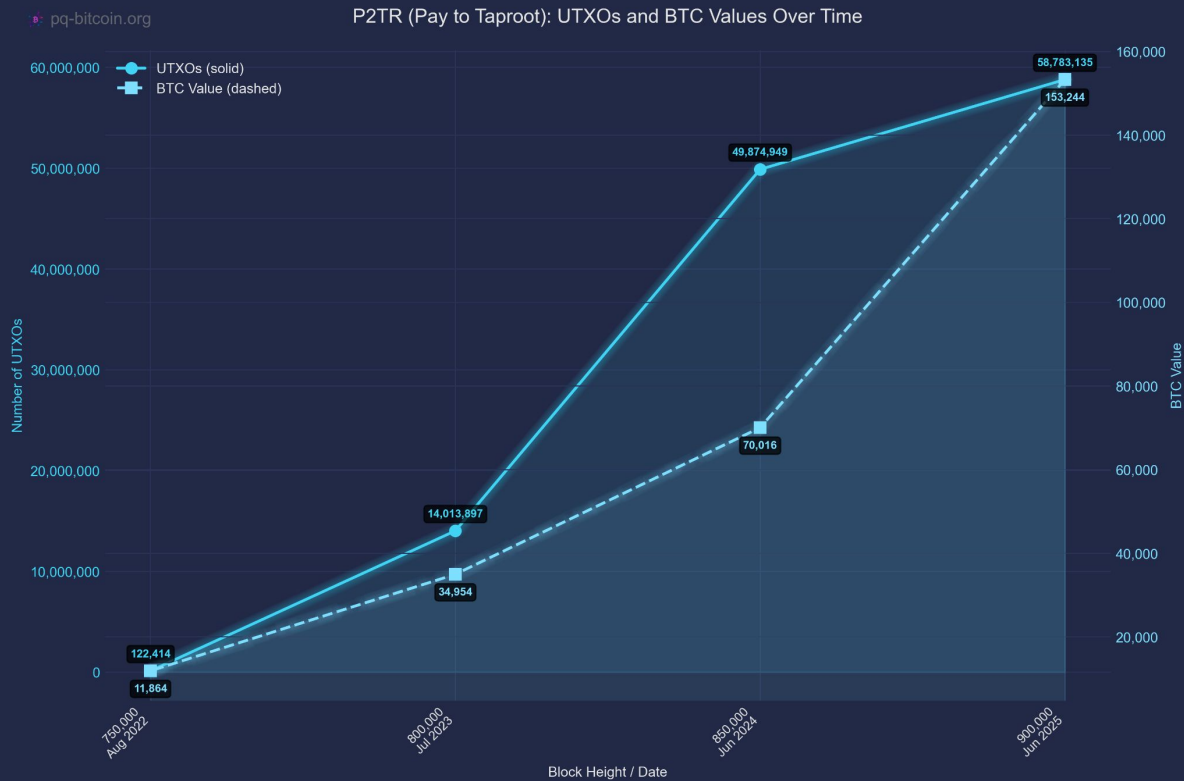
INHERENTLY VULNERABLE SCRIPTS - P2MS

New UTXOs are being created today, but miniscule BTC value

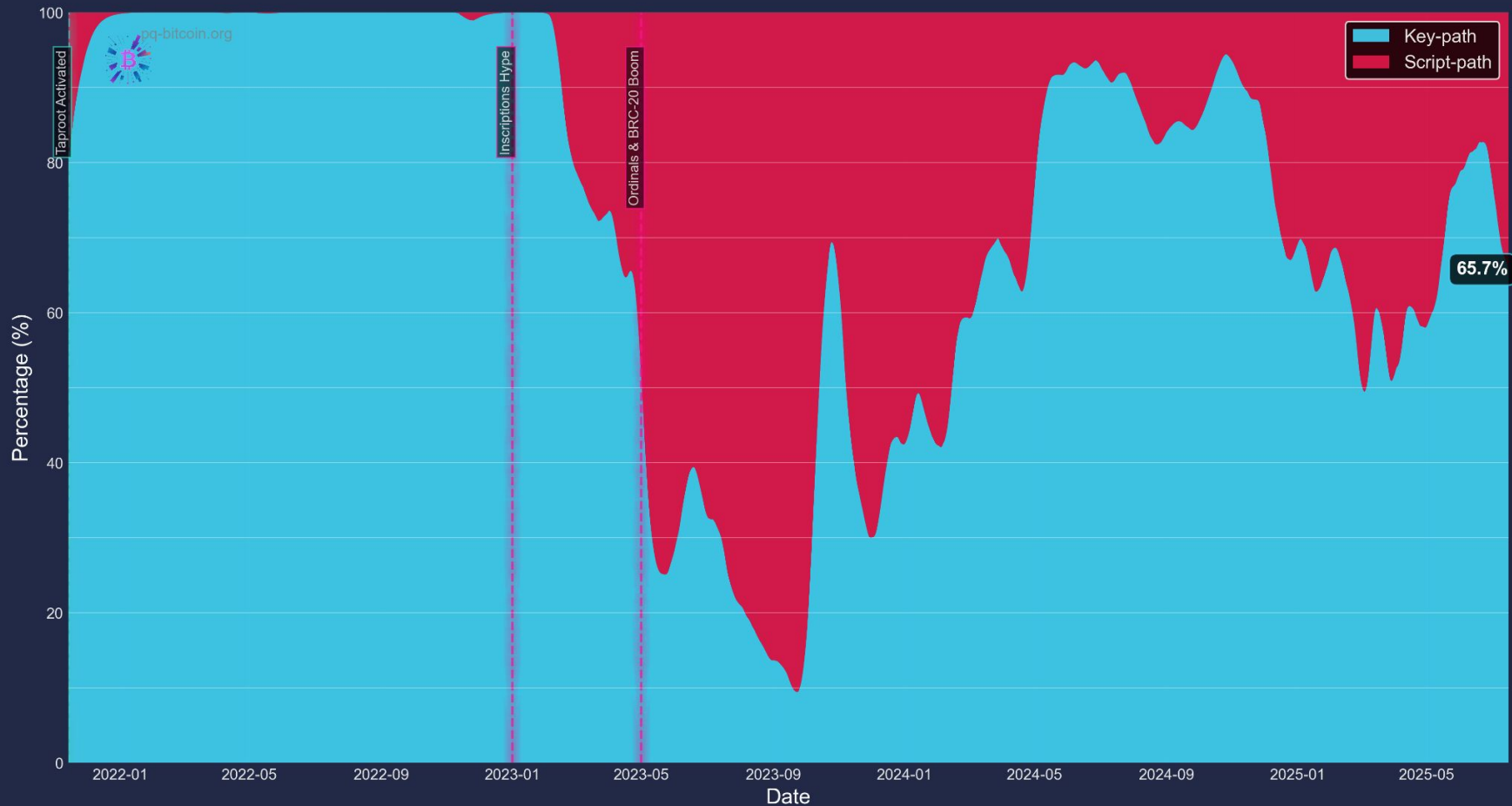


INHERENTLY VULNERABLE SCRIPTS - P2TR

Growing in both # of UTXOs and BTC encumbered



P2TR (Pay to Taproot): Key-path vs Script-path



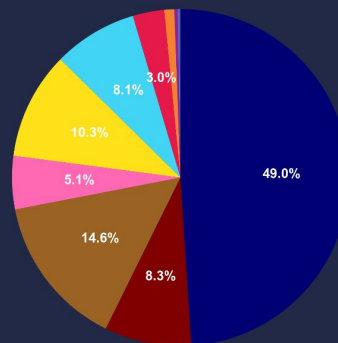
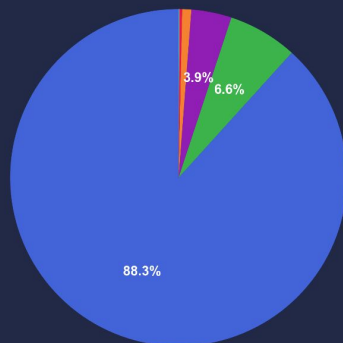


pq-bitcoin.org

Count

P2TR UTXO Distribution by Size Range

Value



P2TR Distribution Summary

Size Range	UTXO Count	Count %	Value (BTC)	Value %
0 - 1k sats	51,218,283	88.3%	248.9	0.2%
1k - 10k sats	3,855,459	6.6%	114.3	0.1%
10k - 100k sats	2,242,806	3.9%	545.8	0.3%
0.001 - 0.01 BTC	505,716	0.9%	1.5k	1.0%
0.01 - 0.1 BTC	145,261	0.3%	4.7k	3.0%
0.1 - 1.0 BTC	40,064	0.1%	12.6k	8.1%
1 - 5 BTC	8,029	0.0%	16.1k	10.3%
5 - 10 BTC	1,169	0.0%	8.0k	5.1%
10 - 50 BTC	1,223	0.0%	22.9k	14.6%
50 - 100 BTC	185	0.0%	13.0k	8.3%
> 100 BTC	187	0.0%	76.6k	49.0%

Total P2TR UTXOs: 58,018,382
Total P2TR BTC: 156,336.63



Adapted from [UTXO Set Report](#) by orangesurf

IMPLICATIONS FOR MIGRATION, BURN/STEAL

P2PK - The main problem (we all knew this)

- ~1.72M BTC, 8.65% of total current BTC supply
- *Not idolising, but... “Lost coins only make everyone else's coins worth slightly more. Think of it as a donation to everyone.”*

P2MS - Recent resurgence in use is for STAMPS protocol ([ref](#), [ref](#))

P2TR - A secondary issue

- Small but growing share of value
- Generally key-path spent → BIP-360?

KEY OBSERVATIONS

~6.51M BTC (32.73% of total supply) is immediately QV

Address Reuse: 4.49M BTC → most of these funds can likely be moved? 

- T1000 show higher rates of reuse (30%) than all addresses (15-20%)
- T1000 account for ~2.73M BTC (

Fork exposure: 0.15M BTC → most of these funds can likely be moved? 

P2PK: 1.72M BTC, almost entirely coinbase TX → most likely won't move 

P2TR: 0.15M BTC, many non-economic UTXOs, key-path favoured

Est: <2M BTC of the immediately QV funds can't/won't be moved

THANKS!
&
QUESTIONS?